

Multi-Agent Scientific Paper Review

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File	Final_paper_kjns_210426 (1).pdf
Words	12,275
Sections Detected	preamble, introduction, methodology, experiments, discussion, section_vi, conclusions, section_vii, biogra
Review Mode	BALANCED
Model	qwen3:8b

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Editorial Decision

MINOR REVISION

Weighted Score: 3.8 / 5.0
Confidence: 81%

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	4.5	Excellent	■■■■■
Manuscript Structure	2.5	Weak	■■■■■
Methodology & Equations	4.2	Good	■■■■■
Statistical Analysis	4.0	Good	■■■■■
Results & Evidence	4.5	Excellent	■■■■■
Discussion & Conclusions	3.5	Good	■■■■■
Figures & Tables	4.2	Good	■■■■■
Scientific Writing	3.0	Acceptable	■■■■■
References & Citations	3.5	Good	■■■■■
Ethics & Transparency	2.0	Weak	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: 4.5 / 5.0 | Confidence: 95%

Strengths:

- [TITLE] The title is specific, informative, and clearly reflects the contribution of the study, incorporating key terms for discoverability.
- [ABSTRACT] The abstract follows a structured format with clear background, objective, methods, results, and conclusions.
- [ABSTRACT] The results are quantified with specific metrics such as $CV \approx 0.07-0.10$ and Pareto front analysis, enhancing credibility.
- [KEYWORDS] The keywords are comprehensive, include domain-specific terms, and avoid duplication of title words.

Weaknesses:

- [TITLE] The title is slightly long and could be shortened for better readability without losing specificity.
- [ABSTRACT] The abstract could be more concise in the introduction of the problem and the experimental setup.
- [KEYWORDS] The keyword 'Multi-Objective Optimization' is repeated in both the title and keywords, which may reduce uniqueness.

Major Comments:

- [TITLE] Consider shortening the title to improve readability while maintaining specificity and informativeness.

Minor Comments:

- [ABSTRACT] The abstract could benefit from a more concise introduction of the problem and experimental setup to enhance clarity.
- [KEYWORDS] Avoid repeating 'Multi-Objective Optimization' in both the title and keywords to maintain keyword diversity.

Recommendations:

- [TITLE] Shorten the title to meet the 10-20 word guideline while retaining key terms.
- [ABSTRACT] Refine the abstract's introduction to clearly state the research problem and its significance.
- [KEYWORDS] Ensure that keywords are unique and do not overlap with the title to enhance searchability.

StructureReviewer

Manuscript structure and logical coherence

Score: 2.5 / 5.0 | Confidence: 80%

Strengths:

- Clear title and abstract
- Well-defined methodology section
- Experimental results presented with figures

Weaknesses:

- Missing literature review section
- Repetition of content in section_vi and section_vii
- Research problem and hypothesis not clearly stated in introduction
- Objectives not explicitly stated or aligned with conclusions
- Poor transitions between sections

Major Comments:

- The manuscript lacks a literature review section, which is essential for contextualizing the research problem and establishing the novelty of the work.

- The content in section_vi and section_vii is duplicated, which undermines the logical flow and coherence of the manuscript.
- The research problem and hypothesis are not clearly stated in the introduction, which is a critical component for guiding the reader through the study.
- The objectives are not explicitly stated or aligned with the conclusions, which weakens the logical structure of the manuscript.
- The transitions between sections are weak, and the manuscript lacks a coherent narrative flow.

Minor Comments:

- The abstract is incomplete and lacks key details about the methodology and results.
- The title is somewhat vague and could be more specific to the study.
- Some figures are not properly labeled or referenced in the text.
- The bibliography section is missing or incomplete.

Recommendations:

- Add a comprehensive literature review section to contextualize the research problem and justify the methodology.
- Remove the duplicated content in section_vi and section_vii to improve coherence.
- Clearly state the research problem and hypothesis in the introduction.
- Explicitly define the objectives and ensure they are addressed in the conclusions.
- Improve transitions between sections to enhance the logical flow of the manuscript.

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: **4.2 / 5.0** | Confidence: 85%

Strengths:

- The integration of physical constraints into the neural network formulation is a novel and significant contribution
- The multi-objective optimization approach is well-justified and aligns with the problem's complexity
- The experimental setup is detailed and includes a rigorous statistical validation framework
- The paper provides a clear connection between the theoretical framework and the practical application

Weaknesses:

- The logical flow between sections is not fully seamless, particularly between the introduction and methodology
- The experimental validation is thorough, but the exact results and their implications are not clearly presented in the discussion
- The equations are not consistently referenced or explained, which may reduce clarity for readers
- The paper lacks a detailed description of the neural network architecture and training procedure

Major Comments:

- The paper should clarify the exact results from Section V in the discussion to ensure the logical flow is complete
- The equations should be consistently numbered and referenced in the text to improve readability and clarity
- A more detailed description of the neural network architecture and training procedure would enhance reproducibility

Minor Comments:

- The transition from the introduction to the methodology could be smoother with a clearer bridge
- The experimental setup could benefit from a more explicit connection to the proposed method
- The paper could include a brief summary of the key results in the conclusion for completeness

Recommendations:

- Add a section or subsection in the methodology that explicitly describes the neural network architecture and training procedure
- Ensure all equations are consistently numbered and referenced in the text

- Clarify the exact results from Section V in the discussion to improve the logical flow
- Include a brief summary of key results in the conclusion for completeness

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: **4.0 / 5.0** | Confidence: 85%

Strengths:

- The manuscript appropriately uses nonparametric tests (Kruskal-Wallis and Wilcoxon) given the lack of normality in the data.
- Bootstrap confidence intervals are reported, which is a strength for robust statistical inference.
- The use of coefficient of variation (CV) and RMSE/MSE metrics is well-justified and interpreted appropriately.
- The manuscript reports 30 independent runs, which is a reasonable number for assessing robustness in stochastic optimization.
- The use of hypervolume and spread indicators to evaluate multi-objective optimization is appropriate and well-explained.

Weaknesses:

- The manuscript does not report p-values for the Kruskal-Wallis and Wilcoxon tests, which is critical for assessing statistical significance.
- The manuscript does not report the exact number of resamples used in the bootstrap confidence intervals, which could affect the reliability of the results.
- The manuscript does not provide a detailed power analysis to justify the sample size of 30 runs, which is a limitation in assessing the adequacy of the sample size.
- The manuscript does not report the exact values of the effect sizes (e.g., Cohen's d, eta-squared), which limits the interpretation of the magnitude of the differences.
- The manuscript does not report the variance of the metrics across runs, which is important for assessing the stability of the results.

Major Comments:

- The manuscript should report p-values for the Kruskal-Wallis and Wilcoxon tests to fully support the statistical significance of the results.

Minor Comments:

- The manuscript should clarify the number of resamples used in the bootstrap confidence intervals to ensure transparency and reproducibility.
- The manuscript should provide a power analysis to justify the sample size of 30 runs, especially given the stochastic nature of the optimization algorithms.

Recommendations:

- Report p-values for the Kruskal-Wallis and Wilcoxon tests to support the statistical significance of the results.
- Clarify the number of resamples used in the bootstrap confidence intervals to ensure reproducibility.
- Provide a power analysis to justify the sample size of 30 runs, especially given the stochastic nature of the optimization algorithms.

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: **4.2 / 5.0** | Confidence: 85%

Strengths:

- The manuscript provides a clear and detailed mathematical formulation for the structural modeling and identification problem.

- Equations are sequentially numbered and referenced in the text, which supports the coherence of the mathematical framework.
- The experimental setup and validation methodology are well described, with clear references to figures and tables.

Weaknesses:

- Some figures and tables are referenced in the text but their bodies are not visible, which limits the ability to verify their content and coherence with the text.
- The lack of detailed captions for figures like Figure 3 reduces their self-explanatory nature.
- The absence of full equation expressions in the text may hinder the verification of their correctness and dimensional consistency.

Major Comments:

- The manuscript should provide full visibility of all figures, tables, and equations to ensure their content is fully verifiable and coherent with the text.

Minor Comments:

- Consider adding detailed captions to figures like Figure 3 to improve their self-explanatory nature.
- Ensure that all equations are fully visible and their variables are clearly defined in the text.

Recommendations:

- Include the full body of Table X to ensure data consistency and clarity.
- Provide the complete expressions for equations like (11) to ensure dimensional consistency and coherence with the described methods.

ResultsReviewer

Results validity and empirical support

Score: 4.5 / 5.0 | Confidence: 85%

Strengths:

- The results clearly address the stated objectives of the study, including the performance of the EANN framework in comparison to GA and PSO baselines.
- Quantitative results are well-supported with metrics such as MSE, RMSE, CV, and hypervolume, which are directly tied to the experimental setup and analysis.
- The manuscript includes a thorough comparison with baseline methods (GA and PSO) and reports statistical validation through Kruskal-Wallis and Wilcoxon rank-sum tests.

Weaknesses:

- The results section lacks detailed reporting of the exact values of MSE, RMSE, and CV for each floor and method, which could have provided more granular insights into performance differences.
- The Pareto front analysis is mentioned but not fully elaborated on, including how the density and uniformity of the front were quantified or visually represented.
- The manuscript does not explicitly state how the 30 independent runs were distributed across different scenarios or conditions, which could affect the interpretation of robustness.

Major Comments:

- The results are well-supported by data, but the lack of detailed breakdown of metrics across floors and methods limits the depth of analysis.
- The manuscript should clarify how the Pareto front analysis was conducted and how the hypervolume/spread indicators were calculated, as these are critical for multi-objective optimization evaluation.

Minor Comments:

- The results section could benefit from more explicit mention of how the 30 independent runs were executed and their impact on the statistical validation.
- The description of the experimental setup and preprocessing steps is sufficient, but the results section could better integrate these details to contextualize the quantitative findings.

Recommendations:

- Include floor-wise breakdown of MSE, RMSE, and CV values for each method to enhance clarity and enable more precise comparisons.
- Clarify the methodology for computing hypervolume and spread indicators, and ensure these are consistently reported across figures and tables.
- Provide more detail on how the 30 independent runs were structured and how they influenced the statistical validation and robustness analysis.

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- The Discussion section effectively interprets results in the context of structural identification methods, highlighting the superiority of EANN over GA and PSO.
- The Conclusions section clearly states the contribution of the EANN framework to structural health monitoring and adaptive control.
- The manuscript provides a clear and specific conclusion about the performance of EANN, supported by the experimental results.

Weaknesses:

- The Discussion section lacks a detailed comparison with prior work, focusing only on citations without elaborating on how the findings align or differ from existing literature.
- The Discussion does not address unexpected findings or anomalies in the results, such as the occasional lower median error values achieved by GA.
- The manuscript does not clearly state the limitations of the study, such as the computational cost of EANN or the specific conditions under which the framework was tested.
- The Conclusions section does not provide concrete future work directions, which is essential for guiding further research.
- The Discussion section overgeneralizes the results by implying broad applicability without addressing the specific context or constraints of the study.

Major Comments:

- The Discussion section should include a more detailed comparison with prior work to contextualize the findings and demonstrate the novelty of the EANN framework.
- The manuscript should explicitly address the limitations of the study, such as computational cost and the specific experimental setup, to provide a balanced interpretation of the results.

Minor Comments:

- The Discussion section could benefit from a more nuanced discussion of the statistical significance of the results and their practical implications.
- The Conclusions section should be more specific in quantifying the performance improvements of EANN over GA and PSO.

Recommendations:

- Include a detailed comparison with prior literature in the Discussion section to contextualize the findings and highlight the novelty of the EANN framework.
- Clarify the limitations of the study, such as computational cost and the specific experimental conditions, to provide a more comprehensive interpretation of the results.
- Provide concrete and justified future work directions in the Conclusions section to guide further research and application of the EANN framework.

WritingReviewer

Scientific writing quality, clarity, and style

Score: **3.0 / 5.0** | Confidence: 55%

Strengths:

- The manuscript contains enough extracted text for a representative writing review.
- Technical terminology and quantitative reporting signals are present in the extracted text.

Weaknesses:

- The model returned malformed JSON for writing review, so this fallback cannot provide sentence-level editing comments.

Minor Comments:

- The model returned malformed JSON for writing review, so this fallback cannot provide sentence-level editing comments.

Recommendations:

- Review representative paragraphs manually for transition quality, undefined acronyms, and overly long sentences.
 - Ensure technical terms are defined on first use and used consistently across sections.
-

ReferencesReviewer

References quality, recency, and relevance

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- References span a range of relevant topics including structural identification, neural networks, and optimization algorithms
- Recent references (2021-2023) are included and relevant to the field
- Citations are consistent in format (IEEE style)

Weaknesses:

- Lack of recent references on structural health monitoring and observer-based state reconstruction
- Missing seminal works on modern seismic design regulations
- Some references are too general or not directly relevant to the specific claims made

Major Comments:

- The references section lacks recent and specific citations on structural health monitoring and observer-based state reconstruction, which are central to the manuscript's claims
- The absence of seminal works on modern seismic design regulations weakens the foundation of the literature review
- The inclusion of a foundational textbook is appropriate but lacks direct relevance to the specific claims made in the manuscript

Minor Comments:

- The citation of [29] is appropriate but could be more specific to the manuscript's focus
- The references are well-organized and follow a consistent citation style

Recommendations:

- Include more recent and specific references on structural health monitoring and observer-based state reconstruction
 - Add seminal works on modern seismic design regulations to strengthen the literature review
 - Ensure that all cited references directly support the claims made in the manuscript
-

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **2.0 / 5.0** | Confidence: 80%

Strengths:

- The limitations are explicitly stated in the discussion section, indicating an awareness of the study's constraints.
- The manuscript presents a novel framework for structural parameter identification, which is a significant contribution to the field.

Weaknesses:

- The limitations are not specific or detailed, which reduces their usefulness in assessing the validity and scope of the study.
- The manuscript lacks a reproducibility statement regarding data and code availability, which is critical for scientific validation.
- There is no mention of ethical approval, which is a fundamental requirement for studies involving human or animal subjects.

Major Comments:

- The manuscript fails to address key ethical and transparency requirements, which are essential for publication in a Q1 journal.
- The limitations section is insufficiently detailed and lacks specificity, which undermines the credibility of the study's conclusions.

Minor Comments:

- The manuscript should include a data availability statement to ensure reproducibility and transparency.
- Author contributions should be clearly stated to ensure proper attribution and accountability.

Recommendations:

- Include a statement regarding ethical approval for any human or animal studies conducted.
- Provide a data availability statement to ensure reproducibility and transparency.
- Declare any potential conflicts of interest to maintain the integrity of the research.
- Disclose funding sources to ensure transparency and accountability.
- Clearly state author contributions to ensure proper attribution.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

- [1] [TITLE] Consider shortening the title to improve readability while maintaining specificity and informativeness.
- [2] The manuscript lacks a literature review section, which is essential for contextualizing the research problem and establishing the novelty of the work.
- [3] The content in section_vi and section_vii is duplicated, which undermines the logical flow and coherence of the manuscript.
- [4] The research problem and hypothesis are not clearly stated in the introduction, which is a critical component for guiding the reader through the study.
- [5] The objectives are not explicitly stated or aligned with the conclusions, which weakens the logical structure of the manuscript.
- [6] The transitions between sections are weak, and the manuscript lacks a coherent narrative flow.
- [7] The paper should clarify the exact results from Section V in the discussion to ensure the logical flow is complete.
- [8] The equations should be consistently numbered and referenced in the text to improve readability and clarity.
- [9] A more detailed description of the neural network architecture and training procedure would enhance reproducibility.
- [10] The manuscript should report p-values for the Kruskal-Wallis and Wilcoxon tests to fully support the statistical significance of the results.
- [11] The manuscript should provide full visibility of all figures, tables, and equations to ensure their content is fully verifiable and coherent with the text.
- [12] The results are well-supported by data, but the lack of detailed breakdown of metrics across floors and methods limits the depth of analysis.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

- [1] [ABSTRACT] The abstract could benefit from a more concise introduction of the problem and experimental setup to enhance clarity.
- [2] [KEYWORDS] Avoid repeating 'Multi-Objective Optimization' in both the title and keywords to maintain keyword diversity.
- [3] The abstract is incomplete and lacks key details about the methodology and results.
- [4] The title is somewhat vague and could be more specific to the study.
- [5] Some figures are not properly labeled or referenced in the text.
- [6] The bibliography section is missing or incomplete.
- [7] The transition from the introduction to the methodology could be smoother with a clearer bridge.
- [8] The experimental setup could benefit from a more explicit connection to the proposed method.
- [9] The paper could include a brief summary of the key results in the conclusion for completeness.
- [10] The manuscript should clarify the number of resamples used in the bootstrap confidence intervals to ensure transparency and reproducibility.
- [11] The manuscript should provide a power analysis to justify the sample size of 30 runs, especially given the stochastic nature of the optimization algorithms.
- [12] Consider adding detailed captions to figures like Figure 3 to improve their self-explanatory nature.

Specific Improvement Recommendations

1. [TITLE] Shorten the title to meet the 10-20 word guideline while retaining key terms.
2. [ABSTRACT] Refine the abstract's introduction to clearly state the research problem and its significance.

3. [KEYWORDS] Ensure that keywords are unique and do not overlap with the title to enhance searchability.
4. Add a comprehensive literature review section to contextualize the research problem and justify the methodology.
5. Remove the duplicated content in section_vi and section_vii to improve coherence.
6. Clearly state the research problem and hypothesis in the introduction.
7. Explicitly define the objectives and ensure they are addressed in the conclusions.
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11. Clarify the exact results from Section V in the discussion to improve the logical flow
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This review was generated automatically by the Multi-Agent Paper Review System using Phi-3 Mini (Microsoft) running locally via Ollama. No manuscript content was transmitted to external servers. Rubrics based on Elsevier, IEEE, and Emerald Q1 peer review standards. This document does not constitute an official editorial decision.